

1 Instruction Set

The virtual machine works with integers. Character values are represented as integers as well (ASCII coding).

Code	Op	Semantics	Stack Change
0	JMP_x	if(pop()!=0) {pc+=x}	-1
1	IS_NEG	push(pop()<0 ? 1 : 0)	±0
2	NOT	push(pop()==0 ? 1 : 0)	±0
3	LOAD_x	push(data[x])	+1
4	STORE_x	data[x] = pop()	-1
5	CONST_x	push(x)	+1
6	DUP	tmp=pop(); push(tmp); push(tmp)	+1
7	POP	pop()	-1
8	ADD	push(pop()+pop())	-1
9	NEG	push(-pop())	±0
10	MUL	push(pop()*pop())	-1
11	DIV	denom = pop(); push(pop()/denom)	-1
12	REM	denom = pop(); push(pop()%denom)	-1
13	IN	push(readChar())	+1
14	OUT	printChar(pop())	-1
15	OUT_INT	printInt(pop())	-1

2 Helper Instructions

Instruction	Replacement
SUB (subtract)	NEG ADD (=a + (-b))
$a < b$	$a - b$ IS_NEG
$a > b$	$b - a$ IS_NEG
$a \leq b$	$a > b$ NOT
$a \geq b$	$a < b$ NOT
$a == b$	$a - b$ NOT
$a \neq b$	$a - b$